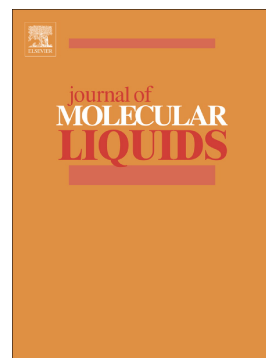


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Self-diffusion in ionic liquids with nitrate anion: effects of confinement between glass plates and static magnetic field

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Abstract

Alkylammonium nitrate protic ionic liquids (ILs), when placed between flat polar borosilicate glass plates, have demonstrated enhanced diffusivity and the gradual decrease of diffusivity after exposure to an external static magnetic field (Filippov *et al.*, *J. Mol. Liq.*, 2018, 268, 49). This phenomenon has been explained by phase transformations taking place in the ILs. In this study, we observed similar processes occurring in systems prepared with ethylammonium nitrate confined between quartz plates. A higher content of silicon oxide in the plates does not significantly alters the phenomenon previously found in the system prepared with borosilicate glass plates. For the first time, we have observed similar effects of confinement and magnetic field on the aprotic IL, 1-ethyl-3-methylimidazolium nitrate. Substitution of the ethylammonium cation with a 1-ethyl-3-methylimidazolium cation slows down the kinetics and increases

magnitude of the processes occurring in the IL exposed to a magnetic field. We suggested that the main factor determining these effects is the presence and modification of the hydrogen-bonding network in the studied protic and aprotic ILs. The process of inverse phase transformation for the confined ethylammonium nitrate after removing the sample from the magnetic field was observed and analysed.

1. Introduction

Ionic liquids (ILs) are a relatively new type of material prepared from organic cations and either organic or inorganic anions that persist in a liquid state near ambient temperature [1-3]. Protic IL is formed by the transfer of a proton from Brønsted acid to Brønsted base, while aprotic cations tend to have C-H bond as the primary H-bond donor unit [2,4]. Structures and properties of ILs at solid-IL interfaces and in confinements are of significant importance to many technological applications of the ILs, including electrochemical processes on the electrodes of batteries and electroplating, in fuel cell membranes, friction surfaces, dissolved solids, *etc.* [5-9]. Therefore, processes near surfaces and in confinements have attracted special interest during the last few years. It has been observed that diffusion coefficients of some nano-confined ILs in aligned carbon nanotubes [10], mesoporous carbon [11] and silicon [12,13] increases by a factor of 2-3 or even more as compared to the diffusivities of ions in the bulk ILs. This has been explained by loose packing of ions in the tight confinement [12]. Our group was the first to revealed variations in local and translational dynamic properties [14] and melting temperatures [15] of ILs confined in micrometer-size range of restrictions. Spontaneous reaction in the aprotic IL 1-ethyl-3-methylimidazolium acetate confined between glass plates, leading to the formation of a neutral chemical moiety, which was assigned as N-heterocyclic carbene has been

observed [16]. Enhanced diffusion of alkylammonium cations has been observed for the protic ILs ethylammonium nitrate (EAN) and propylammonium nitrate (PAN) confined between borosilicate glass plates [14,15]. Magnetic field strength is an important external variable that may influence the phase and dynamic properties of some liquid systems such as liquid crystals with molecules having anisotropic magnetic susceptibility [17] and solid materials, such as organic-based semiconductors [18]. There are reports on magnetic ILs, which contain ferromagnetic ions in their chemical structure, but no perceptible magnetic field effects have been observed for non-magnetic ILs before [3]. Recently it was found that both self-diffusion and NMR relaxation of EAN and PAN confined between borosilicate glass plates with inter-plate distances of *ca.* 4 μm are reversibly altered after sample placement in a static magnetic field [15,19]. It has been demonstrated that the dynamic processes can be described well by the Avrami equation, which is typical for autocatalytic processes [15]. Recently, the same phenomena were observed in systems prepared with surface-treated borosilicate glass plates [20] and for EAN-water mixtures [21]. Underlying physical-chemical mechanisms of the observed phase transformations in the confined ILs remain unclear, as well as the range of conditions and ILs at which these effects can occur. Until now, only the protic ILs, alkylammonium nitrates, demonstrated these effects and the solid support used was borosilicate glass. Therefore, more experimental investigations of the micrometer-size confined ILs are required.

In this work, we investigated processes of phase transformations occurring in confined EAN after placing and removing from a strong static magnetic field. The effect of the solid support was analysed by substitution of borosilicate glass of known chemical composition with quartz glass composed of neat silicon oxide. For the first time, the effects of confinement and magnetic field were observed and analysed for

aprotic IL, 1-ethyl-3-methylimidazolium nitrate, [EMIm][NO₃]. The diffusivity of the studied ILs confined between glass plates and the effects of applying a strong static magnetic field were studied by ¹H NMR.

2. Materials and methods

2.1. Sample preparation

Ethylammonium nitrate (EAN) was synthesised and characterised at Chemistry of Interfaces of Luleå University of Technology, as described previously [14]. The quantity of water in the synthesised ILs was < 0.055 wt% was determined by Karl-Fisher titration (Metrohm 917 Karl Fisher Coloumeter with HYDRANAL reagent). 1-ethyl-3-methylimidazolium nitrate, [EMIm][NO₃] (> 99%) was purchased from Sigma-Aldrich and used without any further purification. Chemical structures of the ionic components of these ILs are shown in Fig. 1. No significant impurities were observed in the studied ILs by using ¹H and ¹³C NMR spectra ([14], Fig. S1 in the Electronic Supporting Information (ESI)).

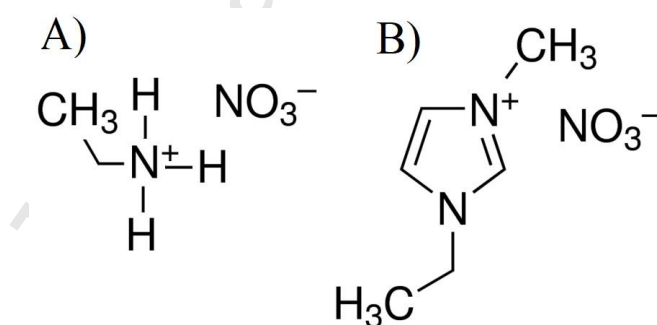


Fig. 1. Chemical structures of **A)** ethylammonium nitrate and **B)** 1-ethyl-3-methylimidazolium nitrate.

Borosilicate glass plates (Thermo Scientific Menzel-Gläser, Menzel GmbH, Germany) and fused quartz glass plates (UEG [Optics] Ltd. The Norman Industrial Estate, Cambridge, GB) with sizes 14 x 2.5 x 0.1 mm were used. The composition of

the borosilicate glass is presented in the ESI. The plates were thoroughly cleaned before sample preparation [14]). Confined ILs were prepared with glass plates settled in a stack (see Fig. S2 in the ESI). A stack of glass plates filled with IL was prepared in a glove box in a dry N₂ atmosphere. Samples were prepared by dropping 2 μ L of the IL onto the first glass plate, placing another glass plate on top, adding 2 μ L of the IL on top of this glass plate, and repeating until the thickness of the stack reached 2.5 mm. IL that overflowed at the edges of the stack was removed by sponging. The sample consisting of a stack of glass plates was positioned in a rectangular glass tube and sealed. A detailed description of the sample preparation and characterisation has been published in our previous papers [14,19]. To allow equilibration of ILs inside the samples, experiments were started one week after the sample preparation.

2.2. Pulsed-field gradient NMR diffusometry

¹H NMR self-diffusion measurements were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a working frequency of 400.27 MHz for ¹H (magnetic field strength of 9.4 T with magnetic field homogeneity better than $4.9 \cdot 10^{-7}$ T) using a Diff50 Pulsed-Field-Gradient (PFG) probe. Homogeneity of the magnetic field was estimated from the ¹H NMR line width. An NMR solenoid ¹H insert was used to macroscopically align the plates of the sample stack at 0 and 90 degrees with respect to the direction of the external magnetic field (the same direction as that of the PFG).

The diffusional decays (DD) were recorded using the spin-echo (SE, at $t_d \leq 5$ ms) or the stimulated echo (StE) pulse trains. For single-component diffusion, the form of the DD can be described as [22-24]:

$$A(\tau, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2}\right) \exp(-\gamma^2 \delta^2 g^2 D t_d) \quad (1a)$$

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right) \exp(-\gamma^2 \delta^2 g^2 D t_d) \quad (1b)$$

for the SE and StE, respectively. Here, A is the integral intensity of the NMR signal, τ and τ_1 are the time intervals in the pulse trains; γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$. D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was $7 \cdot \mu\text{s}$, δ was in the range of $(0.5 \div 2)$ ms, τ was in the range of $(3 \div 5)$ ms, and the amplitude of g was varied from 0.06 up to $29.73 \text{ T} \cdot \text{m}^{-1}$. The recycle delay during accumulation of signal transients was 3.5 s.

As has been shown in our previous work [14], restrictions had no effect on the D of EAN in the direction along the plates, while for diffusion normal to the plates, the effect of the plates on D was noticeable at t_d longer than 3 ms. For diffusion along the plates, D values were obtained by fitting Eqs. (1a) and (1b) to the experimental decays. All measurements were performed at 293 K (room temperature), therefore, we did not have a time gap until the thermal equilibration of the sample in the NMR probe. Measurements were started directly (in $1 \div 2$ min) after placing the sample in the NMR probe, tuning and matching the probe. The time required for a single NMR diffusion experiment was around 30 s. Data were processed using Bruker Topspin 3.5 software. Eqs. (1) and (2) and non-linear least squares regression were used to fit the experimental data to extract D values.

3. Results and discussion

3.1. Diffusivity of EAN between quartz glass plates

Diffusion measurements were performed at 293 K. Diffusion decays of signals corresponding to NH_3 , CH_2 and CH_3 groups were measured as a function of time after placing the sample in the magnetic field of the NMR spectrometer. DDs were always single-exponential (Eqs. 1a and 1b)) and did not depend on the diffusion time. The dependence of the diffusion coefficient on time is shown in Fig. 2, alongside measurements taken earlier for EAN confined between borosilicate glass plates.

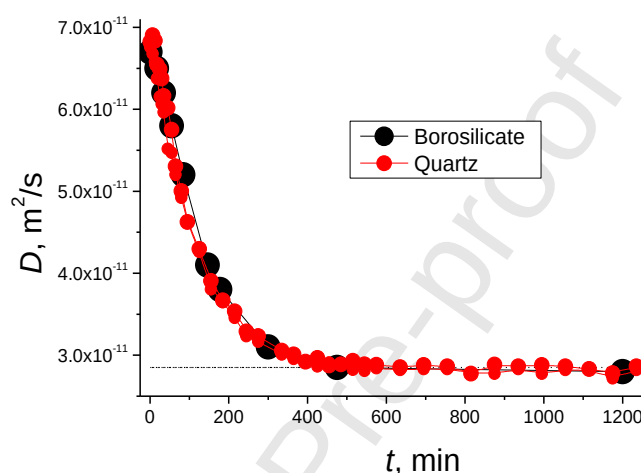


Fig. 2. Change of diffusion coefficient of ethylammonium cations after placing EAN confined between borosilicate glass plates (black larger circles) [19], and quartz plates (red smaller circles) in a magnetic field of 9.4 T. D was measured along the plates, $t_d = 3$ ms. $T = 293$ K.

As seen from Fig. 2, the diffusion coefficient of EAN between quartz plates (red smaller circles) is enhanced to the same degree and decreases at the same rate as in the system with borosilicate plates (black circles) [19]. Therefore, the presence of silicon oxide, but not other chemical compounds, in the glass plates is the factor providing enhanced diffusivity of the confined alkylammonium nitrate and effects of the magnetic field on this confined IL. The presence of other chemical compounds in their fraction $\sim 36\%$ (in the borosilicate glass) does not noticeably influence the observed effects. Use of quartz glass in further experiments may provide an advantage over

borosilicate glass, because quartz glass is more resistant to leaching in aqueous and any other polar media such as ILs [25].

3.2. Diffusivity of confined EAN after removing from a magnetic field

The process of change in the dynamics of alkylammonium nitrates confined between glass plates has been described in detail [14,19]. It has also been shown that the dynamics of EAN reversibly recovered after removing the sample from the magnetic field. However, the kinetics of the recovering process had not been observed and studied yet. Now, we have performed additional experiments with EAN confined between borosilicate glass plates to follow the kinetics of recovery from exposure to a magnetic field. The parameter measured was the diffusion coefficient of the cation $[EA]^+$. Experiments were performed at 293 K twice and with a different time sequence for confidence of the reproducibility of the effect, as described below. Before each experiment, the sample was kept outside of the strong magnetic field for at least one week. Later, the sample was placed in the diffusion probe of the NMR spectrometer (9.4 T). Enhanced diffusivity of the EA cation was registered in the beginning as well as a continuous decrease with time as described earlier [14,19]. In the first experiment, the process of the diffusion coefficient decrease was followed almost to completion (~ 250 min), while in the second experiment, only to a half completion (~ 100 min). Subsequently, the sample was removed from the spectrometer. To trace changes of diffusivity during the process of recovery, the sample was kept outside the magnet and placed in the NMR spectrometer only for the few minutes required for a single measurement. We believe that the limited duration of sample placement in the magnetic field (ca. 2-3 minutes) did not significantly distort the rather slow process of

inverse phase transformation in confined EAN with a characteristic time constant of ca. 500 min. Results of these experiments are shown in Fig. 3.

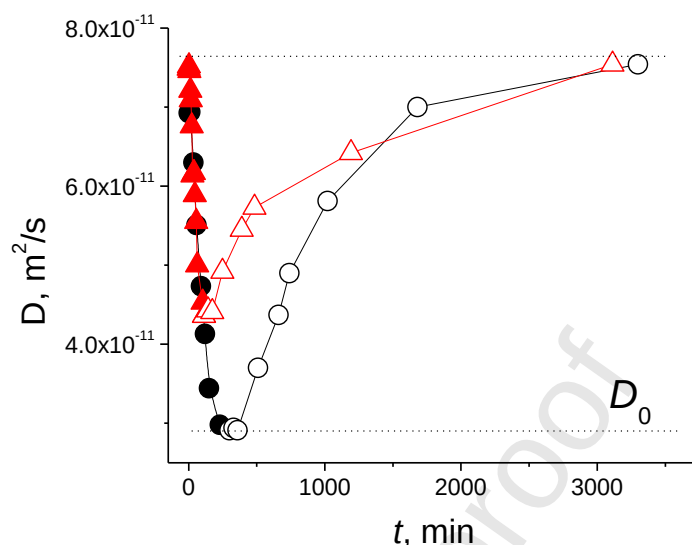


Fig. 3. Change of diffusion coefficient of ethylammonium cations after placing EAN confined between borosilicate glass plates in a magnetic field of 9.4 T (solid symbols) and after removal of the sample from the magnetic field (open symbols). D was measured along the plates, $t_d = 3$ ms. D_0 is the diffusion coefficient of $[EA]^+$ cations in bulk EAN.

As seen from Fig. 3, the process of recovery started just after removing the sample from the strong magnetic field, without a lag. The curves demonstrate typical saturated behaviour. At the end of the process, D reaches the value that was measured at $t = 0$. The kinetics of the process was analysed using Avrami theory [26] and compared to that obtained for processes of change of D of the sample placed in the magnetic field (solid circles in Fig. 3). Details of the analysis can be found in the ESI, Figs. S4-S6. Generally, the kinetics of the reverse phase transformation is well-described by the Avrami formula. The Avrami exponent n is close to 1, the same as was observed for direct transformation [15]. The work of Yang *et al.* [27] showed that homogeneous as

well as heterogeneous nucleation in the surface-independent and diffusion-independent deposition rates give $n \sim 1$ in a 3-dimensional system. This suggests the formation of a primary nucleus of a new phase spontaneously or on heterogeneous nucleation agents in the beginning of the process and further growth of the new phase by deposition of molecules from the original phase on the nucleus surface. Based on the Avrami analysis, we can suggest that mechanisms of reverse phase transformations are the same as the mechanism of direct transformation.

3.3. Diffusivity of 1-ethyl-3-methylimidazolium nitrate confined between glass plates

The ^1H NMR spectrum of $[\text{EMIm}][\text{NO}_3]$ is shown in Fig. S1 in the ESI. All lines in the spectrum belong to the $[\text{EMIm}]^+$ cation. Assignments of the cations' ^1H NMR resonance lines have been done earlier for the ionic liquid 1-ethyl-3-methylimidazolium acetate [16,28]. Accurate chemical shifts of protons of the cation for these two ILs, as well as for $[\text{EMIm}][\text{TFSI}]$ [29], are somewhat different and depend on the structure of the anion and other properties of these ILs, such as their densities, which reflect packing of ions and, thus, the ionic strength of the media. Diffusion coefficients of cations in bulk ILs with $[\text{EMIm}]^+$ measured at the same temperature also depend on the structure of the anion of the particular IL [16,28,29]. In this work, diffusion of the $[\text{EMIm}]^+$ cation of $[\text{EMIm}][\text{NO}_3]$ confined between glass plates was measured along the plates. Diffusion decays obtained from integrated intensities of ^1H NMR resonance lines of the $[\text{EMIm}]^+$ cation in bulk and confined between borosilicate glass plates were of single-component forms (Eq.(1)) and did not depend on the diffusion time in the SE and StE NMR experiments. Measurements are presented in Fig. 4 alongside earlier measurements obtained for EAN confined between borosilicate glass plates.

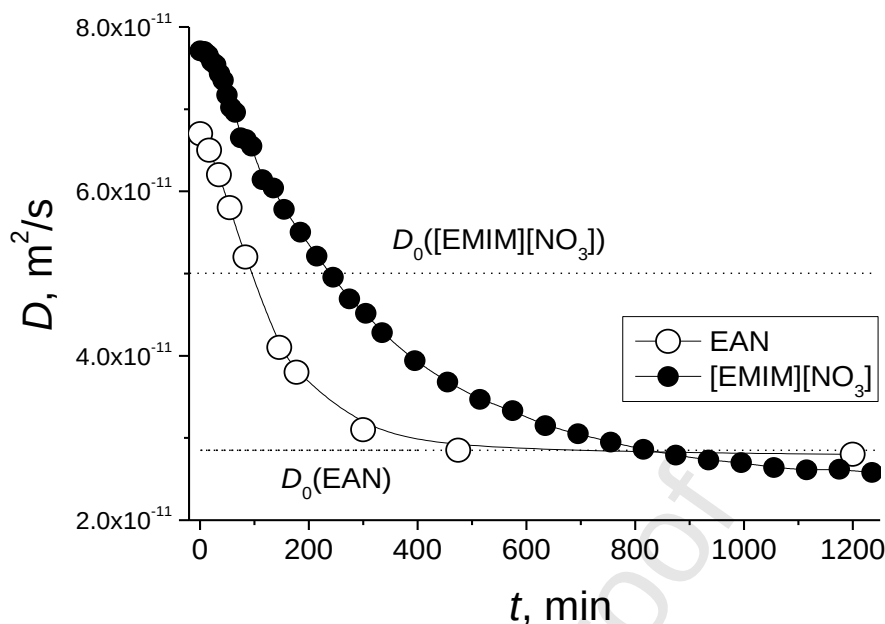


Fig. 4. Change of diffusion coefficient of the EA cation of EAN [19] and [EMIm]⁺ cations of [EMIm][NO₃] IL after placing of the ILs confined between borosilicate glass plates in a magnetic field of 9.4 T. D was measured along the plates, $t_d = 3$ ms. Dotted lines show diffusion coefficients of cations in bulk ILs.

The bulk diffusion coefficient of [EMIm]⁺ in [EMIm][NO₃] is $5.0 \cdot 10^{-11}$ m²/s, which is a factor of ~ 1.7 higher than the diffusion coefficient of [EA]⁺ in bulk EAN (dotted lines in Fig. 4). Confined [EMIm][NO₃] shows enhanced diffusivity, $\sim 7.7 \cdot 10^{-11}$ m²/s, as is the case for EAN ($\sim 6.8 \cdot 10^{-11}$ m²/s). D decreases with time after placement of the sample in the static magnetic field of the NMR spectrometer (solid circles). Comparison of the kinetics occurring in confined [EMIm][NO₃] with that in EAN (open circles) shows several peculiarities of [EMIm][NO₃]. First, the process continues for at least twice the length of time (i. e. asymptotically approaching a new value after $t \sim 1000$ min in comparison to ca. 500 min for EAN). The diffusion coefficient of [EMIm]⁺ decreases from the higher bulk D value to a value a factor of ~ 1.8 lower than

that of bulk D_0 . Overall, D decreases by a factor of ~ 2.9 . For comparison, D of the $[\text{EA}]^+$ cation decreases by a factor of ~ 2.3 . After removing the sample of $[\text{EMIm}][\text{NO}_3]$ from the magnetic field, D is recovering to the initial value and the experiment can be repeated many times, the same as with confined EAN. The kinetics of the decrease in diffusion coefficient of $[\text{EMIm}]^+$ was analysed using the Avrami theory [26,30] in the same manner as we performed before for EAN [15]. Results of this analysis are shown in Fig. S7 in the ESI. By inspecting the curve fits in Fig. S7, one can conclude that the kinetics of transformations in $[\text{EMIm}][\text{NO}_3]$ exposed to a static magnetic field can be described satisfactorily well by the Avrami equation. The Avrami parameter n is related to nucleation. Growth is equal to 1, which corresponds to formation of a primary nucleus of a new phase spontaneously or on heterogeneous nucleation agents in the beginning of the process. Further growth of the new phase occurs by deposition of molecules from the original phase onto the nucleus surface [27].

In our previous works, we observed enhanced diffusivity [14,15] and a decrease of diffusivity of cations for samples placed in a strong, static magnetic field [15,19-21] in confined protic IL alkylammonium nitrates. In this work, for the first time, we observed similar effects for $[\text{EMIm}][\text{NO}_3]$, which is an aprotic IL [31]. In protic ILs, enhancement of diffusivity was accompanied by enhancement in the transverse NMR relaxation rate of exchangeable NH_3 protons, which led to a change in the width of the corresponding line in the ^1H NMR spectrum [15,19]. In the case of $[\text{EMIm}][\text{NO}_3]$, the form of the ^1H NMR spectrum remained unchanged (Fig. S8 in the ESI). Therefore, the studied processes in confined ILs are not related exclusively to the presence of exchangeable protons, changing their mobility in confinement, and being under the influence of a magnetic field. We suggest that the main factor of the observed

processes is the presence of an H-bonding network in the studied protic and aprotic ILs. Alongside ionic interactions, the presence of hydrogen-bonding network affects the physicochemical properties of ILs [31,32]. The strongest hydrogen-bond network is found in protic ILs, where it is highly connected, dense and complex [32]. The H-bonding network is formed in protic IL EAN with proton transfer from the cation to the anion [2]. However, proton transfer does not occur in aprotic [EMIm][NO₃] [31,32], while hydrogen bonds form between protons of CH groups of imidazole and oxygens of [NO₃]⁻ [11]. H-bonds with the aprotic [EMIm]⁺ cation are weaker than those with the protic [C₂H₅NH₃]⁺ cation of EAN. We suggest that the presence of a highly polar silicone oxide on the surface of glass modifies the H-bonding network and changing diffusivity of [EMIm]⁺ in [EMIm][NO₃] with a weaker hydrogen-bonding interactions to a higher degree than for [C₂H₅NH₃]⁺ in EAN. As well as the main factor influencing the dynamic of confined ILs is the modification of hydrogen-bonding network, these effects might be profoundly investigated by FTIR [32,33], Raman spectroscopy [32,34] or obtained from temperature dependence of ¹H NMR chemical shifts [35] to give better understanding of the nature and underlying mechanisms of the effect.

Earlier, we studied the diffusivity and state of 1-ethyl-3-methylimidazolium acetate [EMIm][Ac] confined between borosilicate glass plates [16], which contains the same cation [EMIm]⁺ as in [EMIm][NO₃] studied in this work and also forms an H-bonding network [31,32]. Therefore, one could expect for [EMIm][Ac] the same changes of diffusivity as observed for [EMIm][NO₃]. However, this does not happen; the [EMIm][Ac] IL transforms to a neutral chemical moiety instead, i.e. a 1:1 mixture of neutral N-heterocyclic carbene and protonated acetic acid. This can be due to much weaker H-bonding formed by [EMIm][Ac] in comparison with [EMIm][NO₃] [31], which cannot maintain the integrity of [EMIm][Ac] placed between glass plates.

4. Conclusions

We studied the diffusion of the protic IL ethylammonium nitrate and the aprotic IL 1-ethyl-3-methylimidazolium nitrate ionic liquids placed between glass plates and exposed to a strong, static magnetic field of 9.4 T. Interaction with surface of quartz plates does not significantly changes the dynamics of ethylammonium nitrate in confinement and in a magnetic field compared to that previously found in a system prepared with this IL and borosilicate glass plates, such as enhanced diffusivity in confinement and its gradually decrease in the field. For the first time, we observed similar effects of confinement and magnetic field for the aprotic IL 1-ethyl-3-methylimidazolium nitrate. Now it is evident that the observed phenomena are not directly related to the proton transfer, which occurs in protic ILs. We suggested that the main factor determining these effects is the presence and modification of the hydrogen-bonding networks in the studied protic and aprotic ILs. The process of inverse phase transformation for the confined ethylammonium nitrate after removing the sample from the magnetic field was analysed. Therefore, we have extended the range of ILs and confining materials, which demonstrate anomalies of ion dynamics in confinement and in a magnetic field.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights:

- Enhanced diffusion in ionic liquids observed under confining between quartz plates
- Confinement and static magnetic field can change diffusivity of aprotic ionic liquid
- The process of inverse phase transformation for the confined ionic liquid after removing from the magnetic field was analysed